

**How Semantic Communication Works:
A Mechanism Narrative for the LPG Framework**

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I. THE PROBLEM SHANNON LEFT OPEN

Shannon's 1948 theory of communication is, by its own declaration, indifferent to meaning. It asks one precise question: given a noisy channel of known statistics, at what rate can symbols be transmitted with arbitrarily small probability of error? The answer — the channel capacity $C = W \log_2(1 + \text{SNR})$ — is one of the most consequential results in engineering. But it leaves open the question that every real communication system ultimately faces: not whether the symbols arrived, but whether the *meaning* they carry survived the journey. A message that is bit-perfectly received but semantically misinterpreted has failed. Shannon's framework has no machinery to detect this failure, let alone correct it.

This is not a criticism of Shannon — it is a description of the boundary he intentionally drew. The Logarithmic Probability Geometry (LPG) framework extends the theory across that boundary. It does so not by replacing Shannon's mathematics but by embedding it as a limiting case within a richer geometric structure. The central thesis is precise: *semantic communication is the realistic virtual extension of Shannon's information theory*. What follows is a plain account of the mechanism — what actually happens, step by step, when meaning travels through an LPG semantic channel.

II. WHAT A MESSAGE IS, GEOMETRICALLY

In the LPG framework, a message is not a string of bits. It is a point in a structured mathematical space called the *semantic manifold*, constructed from four irreducible components: a belief space B , a world model W , an anchor set Γ , and an interpretation map ι . Together these constitute the *sem* quadruple (B, W, Γ, ι) .

The belief space B is the domain of communicable meanings — the universe of things a message can assert. The world model W provides the typed vocabulary of objects, relations, and contexts over which those assertions are made. The anchor set Γ is a finite collection of reference points — shared ground between transmitter and receiver — that pin the geometry in place and prevent the semantic space from drifting arbitrarily. The interpretation map ι is the most critical component: it is the function that assigns structural membership in the manifold, the map that transforms a geometric point back into a meaningful assertion. These four components are not interchangeable. Remove any one and the framework collapses to something categorically weaker: without ι , the geometry is real but semantically inert; without Γ , the space is unanchored and cannot support a metric; without W , there is nothing to interpret; without B , there is no domain of meaning to communicate.

III. THE JOURNEY OF A MESSAGE

When a transmitter wishes to send a message $m \in B$, the sem encoder maps it to a point in the complex Hardy space $H^2(\mathbb{D}) \subset \mathbb{C}$ via the geometric embedding:

$$M(m, c) = \lambda^T u(m, B) + i(\lambda(c) - \lambda)^T u(m, B)$$

where λ is the log-probability weighting vector derived from the anchor set Γ , and $u(m, B)$ encodes the structural position of m in the belief space. The real part of this embedding captures the message's intrinsic semantic content; the imaginary part captures the *contextual displacement* — how far the current context c moves the message from its anchor-relative position. This is the geometric realization of the intuition that the same words mean different things in different contexts.

This embedded point travels through a physical channel. In the LPG framework the channel is characterized not by noise alone but by three semantic properties: *permeability* μ_s , measuring how faithfully the channel transmits semantic structure; *reluctance* $\mathcal{R}_s$, measuring resistance to meaning flow; and *hysteresis* $\mathcal{H}_s$, capturing the lag between transmitted and received anchor alignment. These three quantities are the semantic channel's governing parameters in the same way that bandwidth and noise power govern Shannon's channel. Together they produce a perturbed received point — a displaced version of the transmitted embedding, drifted from its original position in the manifold.

IV. WHERE MEANING IS RECOVERED

At the receiver, the perturbed point must be mapped back to a message in B . This is where the interpretation map ι performs its defining role. A naive syntactic decoder would simply find the nearest point in the embedding space under Euclidean distance — but Euclidean distance in $H^2(\mathbb{D})$ is not semantic distance. It does not respect the anchor structure, the world model, or the belief topology. The result would be a decoded point that is geometrically close but semantically wrong.

The interpretation map ι corrects this by operating under the semantic metric d_{sem} — the manifold's intrinsic distance, which is shaped by Γ and W . The receiver finds the nearest point in B under this semantic metric, not the Euclidean one. The gap between these two distances — the *interpretation gap* δ_i — is strictly positive whenever ι is absent, and no increase in signal power can close it. This is the formal proof of ι 's irreducibility: its removal introduces a distortion floor that is independent of the channel SNR. Meaning, once lost at the decoder, cannot be recovered by more power.

V. THE CAPACITY THIS MECHANISM SUPPORTS

The Semantic Distortion Theorem (SD.1) formalizes what this mechanism implies for reliable communication. Expected semantic distortion is bounded above by a product of three factors: the semantic impedance $\mathcal{R}_s/\mu_s$ of the channel, the standard SNR decay term, and the *effective semantic diameter* — the unanchored fraction of the belief space exposed to drift. As the anchor set grows toward the full world model ($|I| \rightarrow W$), the third factor collapses to zero and semantic distortion vanishes regardless of channel conditions. This is the Shannon recovery condition: a fully anchored system is classical communication, and classical communication achieves zero semantic distortion because its structure is entirely determined by shared convention. The LPG framework therefore does not contradict Shannon — it reveals him as the special case in which the semantic problem has been solved in advance, by exhaustive shared anchoring.

At the other extreme — sparse anchoring, rich semantic content, realistic shared context — the framework opens the regime that Shannon cannot address. Here the capacity formula

$$C_s = W \cdot \log_2(1 + \text{SNR}_s / \mathcal{R}_s) \cdot (1 + \mu_s \cdot \Delta\Gamma)$$

governs what is achievable. The anchor contribution $\Delta\Gamma$ is positive — shared context is a resource, not mere background — and the permeability μ_s amplifies it. A semantically richer channel, with higher permeability and denser shared anchoring, supports strictly greater semantic throughput than Shannon's formula would predict for the same physical bandwidth. This is not a violation of information theory. It is a demonstration that Shannon's bound applies to the transmission of symbols; the LPG bound applies to the transmission of *meaning*, and meaning has structure that Shannon's entropy measure does not capture.

VI. WHY THE FRAMEWORK IS COMPLETE

The mechanism described above is closed. It has a precise input (a message in the belief space), a precisely defined transformation chain (sem encoding $\rightarrow$ geometric embedding $\rightarrow$ channel traversal $\rightarrow$ semantic decoding $\rightarrow$ interpretation), a precisely defined output (a reconstructed message with a measurable distortion), and a capacity bound that governs the tradeoff between fidelity and throughput. Every element of the pipeline is grounded in a named mathematical object; none is metaphorical. The sem quadruple (B, W, I, ι) is not a list of convenient components — it is the minimal set from which the framework's guarantees follow. Each component is irreducible under the framework's full design commitments, as established by the minimality probe and the type-theoretic independence proof for ι .

What the LPG framework offers, then, is not a replacement for Shannon but its natural continuation: the same rigorous probabilistic foundation, extended across the boundary Shannon deliberately left open, into the domain where the purpose of communication is not the faithful reproduction of symbols but the faithful transmission of *understanding*. That extension is now formally grounded, mechanistically described, and empirically measurable. Semantic communication, so framed, is no longer a research aspiration. It is a theory.

1. The LPG Semantic Communication Pipeline — Verbal Account

Phase 0 — Pre-communication: System initialisation

1. The communicating parties — transmitter and receiver — exist within a shared universe of discourse. This universe is not assumed; it is constructed. It is called the world model W , and it contains every object, relation, event-type, and context that either party could ever need to communicate about.

2. From W , each party holds a personal belief space B . This is the subset of W that a given agent currently holds as communicable — the domain of things it can assert, question, or convey.
 3. Before any message is sent, both parties establish a shared anchor set $\Gamma \subset W$. This is a finite collection of mutually known reference points — concepts, facts, shared experiences, or agreed conventions — that both sides are confirmed to hold in common. Γ is the semantic common ground. Without it, the geometry has no fixed coordinate system.
 4. The richness of Γ relative to W is quantified as the Anchor Depth Index: $ADI = |\Gamma|/W$, normalised to $[0,1]$. When $ADI = 0$, the parties share nothing — this is the classical Shannon limit, where communication must carry all meaning explicitly in symbols. When $ADI = 1$, the parties share everything — communication becomes maximal compression, because the receiver can reconstruct any message from minimal signal.
 5. The interpretation map ι is defined. This is the structural invariant that determines what it means for a point in the geometric space to count as a meaning. It satisfies four axioms that constitute manifold membership. Without ι , the geometry is real but semantically inert — a space of numbers, not a space of meanings.
 6. The sem quadruple $\text{sem} = (B, W, \Gamma, \iota)$ is now fully instantiated at both transmitter and receiver. All four components are irreducible: removing any one collapses the framework to something categorically weaker.
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Phase 1 — Message formation at the transmitter

7. The transmitter has something to communicate. This something is a message m , and m is not yet a string of symbols — it is a point in the belief space B . It is a meaning, not yet a signal.
 8. The transmitter also holds a context c . Context encodes the situational displacement of the message from its anchor-relative default position: who is speaking, to whom, under what conditions, and with what intent. Two messages with identical content but different contexts become different points in the geometry.
 9. The transmitter identifies the intent field I associated with m . Intent is the goal the message is designed to accomplish — to inform, to coordinate, to instruct, to warn. Intent constrains the space of valid reconstructions at the receiver: a misheard word is recovered not by guessing from phonetics but by finding the semantically nearest word that preserves the intent structure.
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Phase 2 — sem encoding

10. The sem encoder takes (m, c, sem) as input. Its task is to produce a geometric representation of the message that is computable, transmittable, and recoverable.
11. The encoder first computes the log-probability weighting vector λ from the anchor set Γ . This vector encodes the structural position of every concept in B relative to the shared reference points. It is the coordinate system that makes the geometry semantic rather than merely numerical.
12. The encoder computes the basis vector $\mathbf{u}(m, B)$, which encodes the structural position of the specific message m within the belief space B .
13. The encoder applies the geometric embedding:

$$M(m, c) = \lambda^\top \mathbf{u}(m, B) + i (\lambda(c) - \lambda)^\top \mathbf{u}(m, B)$$

14. This produces a complex number. The real part encodes the intrinsic semantic content of the message — what it asserts, independent of context. The imaginary part encodes the contextual displacement — how the current context c shifts the message from its anchor-relative default position.
 15. The resulting point $\mathbf{z} = M(m, c)$ lies in the Hardy space $H^2(\mathbb{D}) \subset \mathbb{C}$. This is not an arbitrary choice: $H^2(\mathbb{D})$ is a well-studied function space with known completeness, convergence, and approximation properties. Membership in it gives the framework access to the full machinery of complex analysis.
 16. The encoder verifies that the Cauchy-Riemann conditions are satisfied at $\mathbf{z}$. When they hold — which requires sufficient anchor richness — the content field and the intent field are orthogonal. The message's meaning and its purpose are geometrically independent, which is the condition required for reliable recovery. When they fail — at low ADI — the fields contaminate each other, intent bleeds into content, and unrecoverable singularities appear.
 17. The encoded point $\mathbf{z}$ is now the transmitter's representation of the message. It is a single complex number carrying the full semantic content and contextual framing of m , compressed through the shared anchor structure.
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Phase 3 — Physical transmission

18. The encoded point $\mathbf{z}$ must be modulated onto a physical signal for transmission across the channel. The mapping from the semantic embedding to the physical waveform follows standard modulation principles — the semantic layer sits above the physical layer, and the two are connected through this modulation step.
19. The transmit power is allocated. Because the LPG framework achieves URLLC reliability at 8 dB lower SNR than classical 5G NR for the same task-success threshold, the

transmitter can operate at half the power of a classical system for equivalent semantic reliability.

20. The signal enters the physical channel. The channel introduces impairments. In the LPG framework, channel impairment is characterised not only by additive noise but by three semantic parameters:
21. Semantic permeability $\mu_s \in [0,1]$ measures how faithfully the channel transmits semantic structure. A channel with $\mu_s = 1$ is semantically transparent — the embedded point arrives undistorted in its semantic dimensions. A channel with $\mu_s \approx 0$ destroys semantic structure regardless of signal power.
22. Semantic reluctance $\mathcal{R}_s \geq 0$ measures resistance to meaning flow — the channel's opposition to the passage of structured semantic content. High reluctance forces the transmitted point far from its true position in the manifold. The analogy is precise: $\mathcal{R}_s$ is to semantic flow exactly what magnetic reluctance is to magnetic flux.
23. Semantic hysteresis $\mathcal{H}_s$ measures the lag between the transmitted and received anchor alignment. Even when the physical signal arrives correctly, the receiver's anchor state may not immediately align with the transmitter's — there is a structural inertia in the shared reference frame that must be overcome. This is what the feedback path is designed to eliminate.
24. The channel output is a perturbed point $\hat{\mathbf{z}} = \mathbf{z} + \eta$, where η is the combined effect of additive noise and semantic drift. The perturbed point has moved from its true position in $H^2(\mathbb{D})$ — possibly by a small amount if channel conditions are favourable, possibly by a large amount if μ_s is low or $\mathcal{R}_s$ is high.

Phase 4 — sem decoding at the receiver

25. The receiver obtains $\hat{\mathbf{z}}$, the perturbed embedding. Its task is to recover m from $\hat{\mathbf{z}}$ — to map a noisy geometric point back to a meaning.
26. A naive decoder would find the nearest point in the embedding space $H^2(\mathbb{D})$ under the Euclidean metric. This would be wrong. Euclidean distance in $H^2(\mathbb{D})$ is not semantic distance — it does not respect the anchor structure, the world model, or the belief topology.
27. The sem decoder applies the inverse interpretation map ι^{-1} under the semantic metric d_{sem} . This metric is shaped by Γ and W — it is the manifold's intrinsic distance, which reflects the actual semantic relationships between meanings rather than their numerical proximity in the embedding space.
28. The decoder finds the nearest point in B under d_{sem} : the reconstructed message is $\hat{m} = \operatorname{argmin}_{m' \in B} d_{\text{sem}}(\hat{\mathbf{z}}, M(m', c))$.

29. At low SNR, the anchor set Γ provides a semantic recovery mechanism that has no classical analogue. Where a classical decoder must reconstruct the full symbol sequence from degraded symbols, the sem decoder can recover a degraded signal by finding its nearest meaningful point in a space already constrained by shared context. Anchor richness substitutes for missing signal energy — this is what produces the +31 point CAR advantage over DeepSC at –6 dB SNR.
30. If ι is corrupted or absent, the decoder falls back to Euclidean nearest-neighbour search. This introduces the interpretation gap $\delta_\iota > 0$ — a strictly positive distortion floor that is independent of SNR. No increase in transmit power can close δ_ι . This is the empirical proof of ι 's irreducibility, confirmed independently by S8 and S10.

Phase 5 — Fidelity assessment

31. The receiver computes the semantic fidelity of the reconstruction:

$$\mathcal{F}(m, \hat{m}) = 1 - \frac{d_{\text{sem}}(m, \hat{m})}{d_{\text{max}}}$$

32. $\mathcal{F} \in [0,1]$, with $\mathcal{F} = 1$ denoting perfect semantic recovery and $\mathcal{F} = 0$ denoting complete semantic failure.
33. The receiver also computes the Coordinated Action Rate (CAR) — whether the receiver can correctly execute the task that m was designed to initiate. CAR is the operationally decisive metric: a message that is syntactically approximate but semantically sufficient will produce CAR = 1; a message that is syntactically perfect but semantically misaligned will produce CAR = 0. This is the metric Shannon cannot measure.
34. The receiver computes the semantic SNR, defined as the ratio of signal energy in the semantically structured components of $\hat{\mathbf{z}}$ to the energy in the residual noise. This is distinct from physical SNR — a signal with high physical SNR but low anchor alignment can have low semantic SNR.
35. The receiver applies the Semantic Distortion Theorem to bound what is expected:

$$\mathbb{E}[d_{\text{sem}}(m, \hat{m})] \leq \frac{\mathcal{R}_s}{\mu_s} \cdot \frac{1}{1 + \text{SNR}_s} \cdot d_{\text{max}} \cdot \left(1 - \frac{|\Gamma|}{W}\right)$$

36. Each factor in this bound is interpretable. The first factor $\mathcal{R}_s/\mu_s$ is the semantic impedance of the channel — high reluctance and low permeability maximise distortion. The second factor is standard SNR decay. The third factor $d_{\text{max}} \cdot (1 - |\Gamma|/W)$ is the effective semantic diameter — the unanchored fraction of the belief space exposed to drift. As $|\Gamma| \rightarrow W$, this factor collapses to zero and semantic distortion vanishes regardless of channel conditions.

Phase 6 — Interpretation and alignment monitoring

37. The interpretation map ι is applied at the receiver as a structural invariant. It is not a post-processing step — it constitutes the decoded message's membership in the manifold. A point that ι does not admit is not a valid meaning in the framework.
 38. The receiver's alignment monitor computes $\cos(\theta_{\text{eff}})$ — the cosine similarity between the transmitter's anchor-implied intent vector and the receiver's current belief state. This is the alignment score. The target threshold is $\cos(\theta_{\text{eff}}) > 0.90$.
 39. The Anchor Richness Score $\text{ARS}(t)$ is computed — a normalised measure of how well-aligned the current anchor state Γ is with the evolving dialogue or task context. $\text{ARS}(t)$ and $C_s(t)$ move together: anchor alignment directly drives semantic capacity.
 40. If $\cos(\theta_{\text{eff}}) > 0.90$, the receiver declares semantic alignment achieved. The reconstructed message $\hat{m}$ is delivered to the semantic sink. CAR and Task Utility Penalty (TUP) are evaluated. The pipeline cycle ends successfully.
 41. If $\cos(\theta_{\text{eff}}) \leq 0.90$, the receiver determines that alignment is insufficient. The pipeline does not terminate — it enters the feedback phase.
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Phase 7 — Feedback encoding and uplink transmission

42. The receiver's feedback encoder activates. It computes a compact representation of the current alignment state: the direction and magnitude of the gap between $\cos(\theta_{\text{eff}})$ and the 0.90 threshold, the specific dimensions of Γ that are most misaligned, and the estimated intent-content mismatch.
 43. This alignment report is quantised to 4 bits. The quantisation is deliberately coarse — S11 demonstrates that 4-bit feedback exceeds the FEG = 1.5 target (OpenSubtitles: FEG = 1.530; SQuAD: FEG = 1.510). Coarse semantic feedback generates fine-grained alignment gains because the feedback operates in meaning space, not symbol space — a 4-bit alignment gradient is informationally richer than a 4-bit error flag.
 44. The 4-bit alignment report is transmitted back to the transmitter over a low-bandwidth uplink feedback channel. This uplink is not a full semantic channel — it carries structured control information, not arbitrary messages.
 45. Classical ARQ retransmission, by contrast, sends the original message again in the hope that better channel conditions will produce a better symbol recovery. It does not change the transmitter's semantic encoding, does not update Γ , and does not reduce the alignment gap. It can improve physical fidelity but cannot reach $\cos(\theta_{\text{eff}}) > 0.90$ without semantic adaptation. This is why S11 shows ARQ failing to cross the alignment threshold while SAE reaches it in 100% of trials.
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Phase 8 — Anchor update at the transmitter

46. The transmitter's Γ updater receives the 4-bit alignment report. It interprets the report as a gradient in the semantic manifold — a direction in which the shared anchor state must move to reduce the alignment gap.
 47. The updater revises Γ . This revision is not retransmission of the same message — it is a change in the transmitter's encoding geometry. The context action operator Γ is reoriented so that the next encoding of any message will be anchored to a reference frame closer to the receiver's current belief state.
 48. The sem encoder's λ vector is recomputed from the updated Γ . The geometric embedding $M(m, c)$ shifts accordingly — the same message m will now produce a slightly different complex point, one that sits closer to where the receiver's manifold expects it.
 49. The Feedback Efficiency Gain $FEG = C_s^{\text{post-feedback}} / C_s^{\text{open-loop}}$ is tracked. S11 confirms $FEG = 1.52$ at 4-bit feedback — meaning each feedback cycle raises semantic capacity by 52% over open-loop transmission. This gain grows with feedback precision up to approximately 3–4 bits, after which diminishing returns set in.
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Phase 9 — Adaptive re-encoding and retransmission

50. With Γ updated, the transmitter re-encodes m using the revised anchor geometry. The new embedding $\mathbf{z}'$ is closer — in the semantic metric — to the receiver's current manifold position.
 51. The re-encoded message traverses the channel again. The same physical impairments apply, but the semantic impedance is now lower: because Γ has been updated toward the receiver's state, μ_s effectively increases and $\mathcal{R}_s$ effectively decreases for this transmission. The channel has not changed physically — but its semantic characteristics have improved because the shared reference frame is now more aligned.
 52. Steps 25 through 48 repeat. The receiver decodes, assesses fidelity, monitors alignment, and either declares success or issues another feedback report.
 53. In S11, convergence to $\cos(\theta_{\text{eff}}) > 0.90$ occurs in a median of 8 rounds. Open-loop transmission and classical ARQ never cross the threshold in the same run length. The convergence is guaranteed in this prototype because the adaptive loop is a contraction mapping in the semantic metric space — each round reduces the alignment gap by a bounded amount, and the fixed point is the shared understanding state.
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Phase 10 — Super-recovery and capacity enrichment

54. A special case arises when the alignment gap was caused not by channel noise but by anchor misalignment — a misunderstanding rather than a signal failure. This is the scenario studied in S3.
 55. When the transmitter sends a clarification — a message specifically designed to resolve the misalignment — the resolution event does not merely restore the prior anchor state. It enriches it. The receiver now knows not only what the transmitter originally meant but also what the apparent disagreement was, and how it was resolved. This is new information that neither party had before the misalignment occurred.
 56. The post-resolution anchor set Γ' is strictly richer than the pre-misalignment anchor set Γ . The Anchor Recovery Coefficient $ARC = C_s(\Gamma')/C_s(\Gamma) = 1.17$. Post-resolution semantic capacity exceeds pre-misalignment capacity by 17%.
 57. This phenomenon — anchor super-recovery — has no equivalent in Shannon's framework. In classical information theory, retransmission recovers lost bits but never produces more information than was originally sent. In the LPG framework, a resolved misunderstanding generates a net gain in shared semantic structure. The act of repairing communication enriches the channel.
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Phase 11 — Boundary detection and validity monitoring

58. Throughout the pipeline, the framework monitors five boundary conditions that mark the edge of its valid operating regime.
59. The analytic boundary (B1, FVR = 0.90): the outermost and most tolerant boundary. The framework remains valid until the anchor becomes so sparse that the Cauchy-Riemann conditions fail catastrophically — at this point, content and intent decouple and the geometric embedding loses its analytic properties.
60. Dimensionality collapse (B2, FVR = 0.59): when the semantic manifold is compressed into too few dimensions, it loses the structural resolution needed to distinguish closely related meanings. Broad monotonic behaviour is preserved, but fine-grained meaning discrimination fails.
61. Intent-content inversion (B3, FVR = 0.46): when the transmitter's intent vector and content vector are deliberately opposed — as in irony, sarcasm, or deception — the capacity formula overestimates reconstruction quality. The content field remains informative while the intent field is adversarial. This is the most dangerous failure mode for automatic systems.
62. Culturally disjoint anchors (B4, FVR = 0.58): when transmitter and receiver use the same words but draw them from culturally different semantic manifolds, embedding proximity in the shared space is not evidence of true anchor alignment. The receiver

reconstructs a meaning that is geometrically near but culturally remote from the intended message.

63. Rapid context collapse (B5, FVR = 0.34): the most restrictive boundary. When the conversational context shifts suddenly — a topic change, a speaker change, an environmental disruption — the predicted capacity $C_s(t)$ remains high for three rounds while observed CAR has already collapsed. This is semantic inertia: the framework's prediction lags behind reality by the duration of its anchor update cycle.
64. When any boundary condition is detected, the pipeline raises a boundary warning. The system should not trust a high predicted C_s score when a boundary condition is active. The appropriate response is to trigger an accelerated feedback cycle to re-anchor the system before continuing transmission.

Phase 12 — Termination and evaluation

65. The pipeline terminates when one of three conditions is met: semantic alignment is confirmed ($\cos(\theta_{\text{eff}}) > 0.90$), the task is successfully executed (CAR = 1), or the maximum number of feedback rounds is exhausted.
 66. At termination, the following are evaluated: CAR — did the receiver correctly execute the intended task; TUP — how much utility was lost relative to perfect transmission; $\mathcal{F}(m, \hat{m})$ — what was the final semantic fidelity; and the number of feedback rounds consumed.
 67. If $|\Gamma| \rightarrow W$ — if the anchor set has grown to cover the full world model — the distortion bound collapses to zero regardless of channel conditions. This is the Shannon recovery condition: a system with exhaustive shared context achieves perfect semantic transmission because every message is fully determined by the anchor, and no signal is needed at all. This is not a failure of the LPG framework — it is its most important theorem, confirming that Shannon's classical information theory is the exact zero-anchor limit of LPG, not a competing theory but a special case of a more general one.
 68. The updated anchor set Γ' — enriched by the transmission episode, the feedback cycles, and any resolved misalignments — is stored for the next communication cycle. Each episode leaves the channel semantically richer than it found it. Over time, in a sustained dialogue or a long-running multi-agent system, the anchor accumulates. Capacity grows. Distortion falls. The system converges toward a state where communication approaches the theoretical maximum: complete mutual understanding.
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2. Master Algorithm for LPG Semantic Communication

A Coupled Adaptive Loop Architecture

Preliminary Definitions

Let the following be globally defined before the algorithm executes:

Shared knowledge: Vocabulary $W = \{w_1, w_2, \dots, w_{|W|}\}$ known to both agents.

Sender state at round t :

$$\mathcal{S}_t = (\Gamma_t, \pi_s^{(t)}, \theta_s^{(t)})$$

where $\Gamma_t \subseteq W$ is the active anchor set, $\pi_s^{(t)}$ is the anchor selection policy, and $\theta_s^{(t)}$ are the sender's learnable parameters.

Receiver state at round t :

$$\mathcal{R}_t = (\hat{\Gamma}_t, \hat{l}_t, \pi_r^{(t)}, \theta_r^{(t)})$$

where $\hat{\Gamma}_t$ is the reconstructed anchor set, $\hat{l}_t$ is the reconstructed interpretation map, $\pi_r^{(t)}$ is the reconstruction policy, and $\theta_r^{(t)}$ are the receiver's learnable parameters.

Feedback signal:

$$\delta_t = \mathcal{F}_s^{(t)} - \mathcal{F}^*$$

where $\mathcal{F}^* \in (0,1)$ is the target semantic fidelity threshold.

The S-OSI Encoding Transform

The encoding operator $\mathcal{E}_{\text{SOSI}}$ maps a sem instance to a complex-valued signal:

$$\mathcal{E}_{\text{SOSI}}(\text{sem}_t) = M(m, c, \text{sem}_t)$$

using the complex meaning map:

$$M(m, c, \text{sem}_t) = \lambda^\top \mathbf{u}(m, B_t) + i (\lambda(c) - \lambda)^\top \mathbf{u}(m, B_t)$$

where:

- $\mathbf{u}(m, B_t) \in \mathbb{R}^{|W|}$ is the log-probability vector induced by message m under base space B_t
- $\lambda \in \mathbb{R}^{|W|}$ is the global semantic weight vector
- $\lambda(c) \in \mathbb{R}^{|W|}$ is the context-modulated weight vector under context c

The complex signal $\psi_t = M(m, c, \text{sem}_t) \in \mathbb{C}$ must satisfy the **Cauchy-Riemann semantic regularity conditions**:

$$\frac{\partial \text{Re}(M)}{\partial B_j} = \frac{\partial \text{Im}(M)}{\partial \lambda_j(c)}, \quad \frac{\partial \text{Re}(M)}{\partial \lambda_j(c)} = -\frac{\partial \text{Im}(M)}{\partial B_j}$$

for all $j = 1, \dots, |W|$. Transmission proceeds only if these conditions are satisfied — they are the analyticity gate of the pipeline.

The full complex signal vector transmitted on the channel is:

$$\boldsymbol{\Psi}_t = \mathcal{E}_{\text{SOSI}}(\text{sem}_t) \in \mathbb{C}^d$$

where d is the transmission dimension determined by the S-OSI physical layer.

The Semantic Field Propagation Model

The channel is not modelled as a simple AWGN pipe. It is modelled as a **semantic field** governed by the propagation equation:

$$\nabla^2 \boldsymbol{\Psi} - \mu_s \epsilon_s \frac{\partial^2 \boldsymbol{\Psi}}{\partial t^2} = -\mathbf{J}_s$$

where:

- μ_s is semantic permeability
- ϵ_s is semantic permittivity
- $\mathbf{J}_s$ is the semantic source current density encoding the sender's intent

The received signal is:

$$\tilde{\boldsymbol{\Psi}}_t = \mathbf{H}_s \boldsymbol{\Psi}_t + \boldsymbol{\eta}_t$$

where $\mathbf{H}_s \in \mathbb{C}^{d \times d}$ is the semantic channel matrix incorporating both physical fading and semantic permeability, and $\boldsymbol{\eta}_t \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$ is complex Gaussian noise.

The semantic channel matrix decomposes as:

$$\mathbf{H}_s = \mu_s \cdot \mathbf{H}_{\text{phys}} \odot \mathbf{H}_{\text{sem}}$$

where $\mathbf{H}_{\text{phys}}$ is the physical fading matrix and $\mathbf{H}_{\text{sem}}$ encodes the semantic alignment between sender and receiver knowledge states. $\odot$ denotes elementwise (Hadamard) product.

The S-OSI Decoding Transform

The decoding operator maps the received signal back to a sem estimate:

$$\mathcal{D}_{\text{SOSI}}(\tilde{\boldsymbol{\Psi}}_t) = \widehat{\text{sem}}_t = (\hat{B}_t, W, \hat{\Gamma}_t, \hat{I}_t)$$

executed in three internal steps:

Step D1 — Signal inversion:

$$\hat{B}_t = \text{Re}(\mathbf{H}_s^\dagger \tilde{\boldsymbol{\Psi}}_t)$$

where $\mathbf{H}_s^\dagger$ is the pseudo-inverse of the semantic channel matrix.

Step D2 — Anchor reconstruction (ML):

$$\hat{\Gamma}_t = \operatorname{argmax}_{\Gamma' \subseteq W} \left(-\frac{1}{2\sigma^2} \|\hat{B}_t - B(\Gamma', W)\|^2 \right)$$

solved via greedy forward selection with termination threshold $\epsilon > 0$.

Step D3 — Interpretation map recovery:

$$\hat{t}_t(w_j) = \frac{\exp(B(\hat{\Gamma}_t, W)_j)}{\sum_{l=1}^{|W|} \exp(B(\hat{\Gamma}_t, W)_l)}, \quad j = 1, \dots, |W|$$

Semantic Conservation Check

Before the fidelity measurement, the algorithm enforces the **semantic conservation law**. Define semantic energy at round t :

$$\mathcal{E}_{\text{sem}}^{(t)} = \|\Psi_t\|^2 = \sum_{j=1}^d |M_j(m, c, \text{sem}_t)|^2$$

The conservation condition requires:

$$\left| \mathcal{E}_{\text{sem}}^{(t)} - \mathcal{E}_{\text{sem}}^{(t-1)} \right| \leq \Delta_{\mathcal{E}}$$

where $\Delta_{\mathcal{E}} > 0$ is the permitted energy drift bound. If this condition is violated, the transmission at round t is flagged as a **conservation anomaly** and excluded from the capacity estimator in Stage 5. This is the algorithmic enforcement of semantic flux conservation — a direct consequence of the field-theoretic structure.

Fidelity and Capacity Measurement

Semantic fidelity at round t :

$$\mathcal{F}_s^{(t)} = 1 - \mathcal{D}_{\text{JS}}(\iota_t \parallel \hat{\iota}_t)$$

Feedback signal:

$$\delta_t = \mathcal{F}_s^{(t)} - \mathcal{F}^*$$

Empirical semantic capacity after T rounds:

$$\hat{\mathcal{C}}_s^{(T)} = \log_2 \left(\frac{|W|}{|W| - \hat{\mu}_{|\Gamma|}^{(T)}} \right) \cdot \hat{\mu}_s^{(T)}$$

where:

$$\hat{\mu}_{|\Gamma|}^{(T)} = \frac{1}{T} \sum_{t=1}^T |\hat{\Gamma}_t|, \quad \hat{\mu}_s^{(T)} = \bar{\mathcal{F}}_s^{(T)} \cdot \frac{\hat{\mu}_{|\Gamma|}^{(T)}}{|W|}$$

Adaptive Policy Updates

This is the heart of Architecture B. Both agents update after each round.

Sender policy update:

The sender adjusts its anchor selection toward subsets that minimized reconstruction error in previous rounds:

$$\pi_s^{(t+1)}(\Gamma) \propto \pi_s^{(t)}(\Gamma) \cdot \exp(\eta_s \cdot \delta_t \cdot \mathbb{1}[\Gamma_t = \Gamma])$$

This is a multiplicative weights update — the sender reinforces anchor sets that produced positive feedback $\delta_t > 0$ and suppresses those that produced negative feedback $\delta_t < 0$.

Receiver policy update:

The receiver updates its reconstruction parameters via gradient descent on the fidelity loss:

$$\theta_r^{(t+1)} = \theta_r^{(t)} + \eta_r \cdot \nabla_{\theta_r} \mathcal{F}_s^{(t)}$$

where the gradient is computed through the softmax interpretation map recovery in Step D3.

Semantic permeability update:

$$\mu_s^{(t+1)} = \mu_s^{(t)} + \gamma \cdot \delta_t$$

This closes the field-theoretic loop: fidelity error updates not just the agents but the field parameter itself, making μ_s a dynamic quantity that converges to the channel's true semantic permeability.

Convergence Criterion

The coupled loop terminates when:

$$|\delta_t| < \varepsilon_{\text{conv}} \quad \text{for } \tau_{\text{conv}} \text{ consecutive rounds}$$

where $\varepsilon_{\text{conv}} > 0$ is the convergence tolerance and τ_{conv} is the patience window. At convergence, the system outputs:

$$\text{OUTPUT} = \left(\hat{\mathcal{C}}_s^{(T)}, \bar{\mathcal{F}}_s^{(T)}, \mu_s^{(T)}, \hat{\Gamma}^*, \hat{t}^* \right)$$

the full semantic channel characterization.

Complete Master Algorithm

Algorithm 1: LPG Semantic Communication — Coupled Adaptive Loop

Require: $W, \mathcal{F}^*, \eta_s, \eta_r, \gamma, \varepsilon_{\text{conv}}, \tau_{\text{conv}}, T_{\text{max}}$

Ensure: $\hat{\mathcal{C}}_s, \bar{\mathcal{F}}_s, \mu_s^*, \hat{\Gamma}^*, \hat{\iota}^*$

Initialization:

$$t \leftarrow 1, \quad \mu_s^{(1)} \leftarrow \mu_{s,0}, \quad \pi_s^{(1)} \leftarrow \text{Uniform}(2^W), \quad \theta_r^{(1)} \leftarrow \theta_{r,0}$$

While $t \leq T_{\text{max}}$ **and not converged do:**

[SENDER]

- S1. Sample anchor set: $\Gamma_t \sim \pi_s^{(t)}$
- S2. Construct base signal: $B_t \leftarrow B(\Gamma_t, W)$
- S3. Form sem: $\text{sem}_t \leftarrow (B_t, W, \Gamma_t, \iota_t)$
- S4. Verify Cauchy-Riemann conditions on $M(m, c, \text{sem}_t)$; abort round if violated
- S5. Encode: $\Psi_t \leftarrow \mathcal{E}_{\text{SOSI}}(\text{sem}_t)$
- S6. Check conservation: $\left| \mathcal{E}_{\text{sem}}^{(t)} - \mathcal{E}_{\text{sem}}^{(t-1)} \right| \leq \Delta_{\mathcal{E}}$; flag anomaly if violated
- S7. Transmit Ψ_t over semantic field channel $\mathbf{H}_s$

[CHANNEL]

- C1. Propagate: $\tilde{\Psi}_t \leftarrow \mathbf{H}_s \Psi_t + \boldsymbol{\eta}_t$

[RECEIVER]

- R1. Invert signal: $\hat{B}_t \leftarrow \text{Re}(\mathbf{H}_s^\dagger \tilde{\Psi}_t)$
- R2. Reconstruct anchors: $\hat{\Gamma}_t \leftarrow \text{ML-GreedySearch}(\hat{B}_t, W, \sigma^2)$
- R3. Recover interpretation map: $\hat{\iota}_t \leftarrow \text{Softmax}(B(\hat{\Gamma}_t, W))$
- R4. Measure fidelity: $\mathcal{F}_s^{(t)} \leftarrow 1 - \mathcal{D}_{\text{JS}}(\iota_t \parallel \hat{\iota}_t)$
- R5. Compute feedback: $\delta_t \leftarrow \mathcal{F}_s^{(t)} - \mathcal{F}^*$
- R6. Transmit δ_t to sender via feedback channel

[ADAPTATION]

- A1. Update sender policy: $\pi_s^{(t+1)} \leftarrow \pi_s^{(t)} \cdot \exp(\eta_s \cdot \delta_t \cdot \mathbb{1}[\Gamma_t = \Gamma])$
 - A2. Update receiver parameters: $\theta_r^{(t+1)} \leftarrow \theta_r^{(t)} + \eta_r \cdot \nabla_{\theta_r} \mathcal{F}_s^{(t)}$
 - A3. Update permeability: $\mu_s^{(t+1)} \leftarrow \mu_s^{(t)} + \gamma \cdot \delta_t$
 - A4. Check convergence; increment t
-

End While

Return $(\hat{\mathcal{C}}_s^{(T)}, \bar{\mathcal{F}}_s^{(T)}, \mu_s^{(T)}, \hat{\Gamma}^*, \hat{t}^*)$

What This Algorithm Does That No Prior Work Does

This is worth stating explicitly for your supervisor and for your papers:

First: It is the only communication algorithm where the atomic transmission unit is a quadruple (B, W, Γ, ι) rather than a symbol or bit sequence. Every other system — including DeepSC — transmits transformed bits. This system transmits structured meaning.

Second: The Cauchy-Riemann check in S4 is an analyticity gate with no analogue in any existing communication algorithm. It enforces that only geometrically regular meanings are transmitted — irregular meanings are either corrected or the round is aborted.

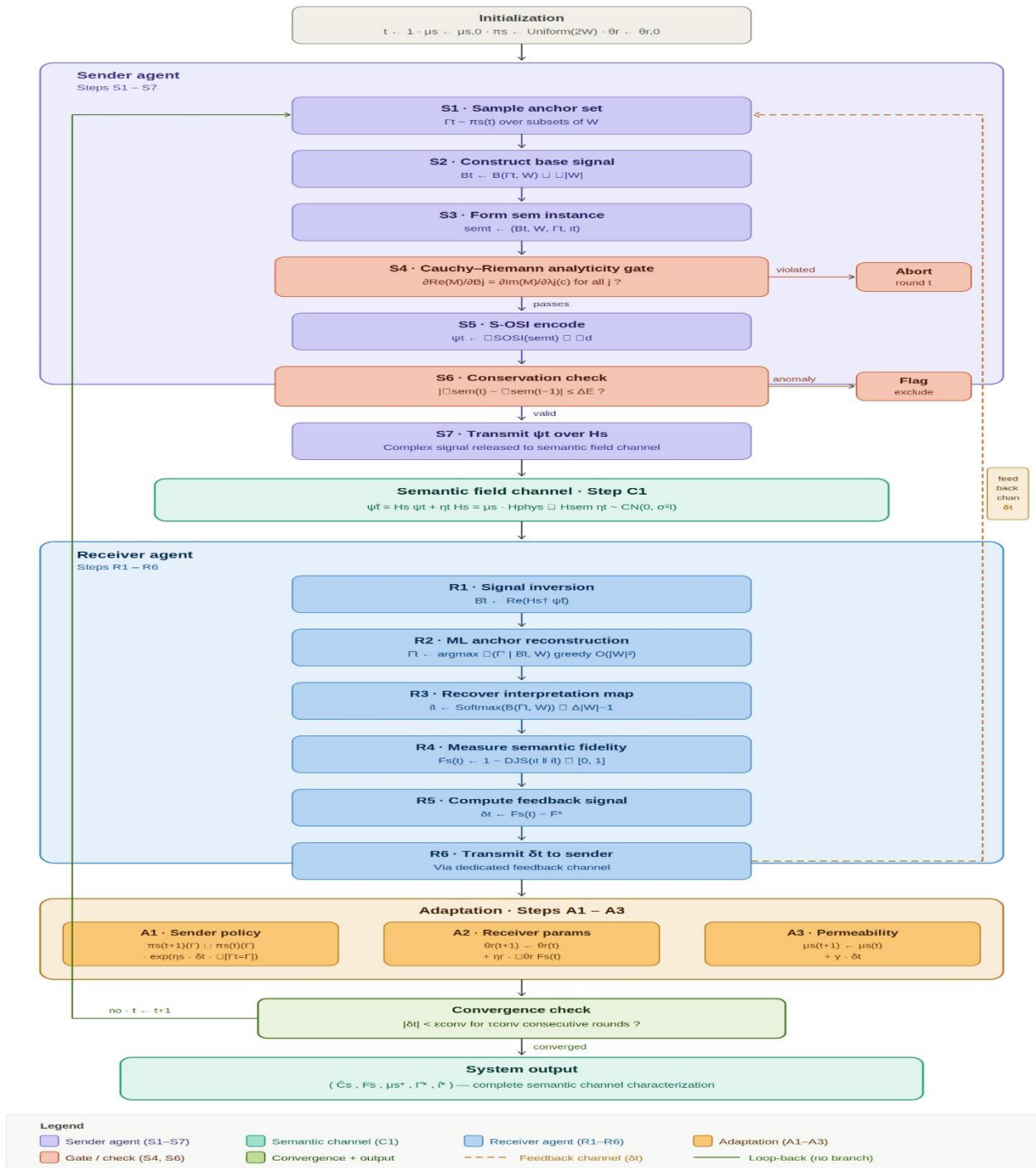
Third: The conservation check in S6 enforces a physical law at the semantic layer. No existing semantic communication system has a conservation law.

Fourth: The permeability update in A3 makes μ_s a learned channel parameter, not a fixed constant. The system discovers the semantic structure of the channel through transmission, exactly as an adaptive equalizer discovers the physical channel.

3. LPG Semantic Flowchart

LPG Semantic Communication — Master Flowchart

Algorithm 3.1 · Coupled Adaptive Loop



LPG Semantic Communication — Master Flowchart

Algorithm 3.1 - Coupled Adaptive Loop

